



ASD810S (Advanced Software Development)

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students at the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to adhering to the institution's policies and rules. The core activity of NUST is learning, and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to complete their own assignments and produce their own academic work, unless a group assignment is given.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works, or inventions of someone else as if they were your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are considered misconduct under the NUST Student Rules and Regulations. Students found guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or expelled. An appropriate endorsement, which will never be erased, is also made on the academic record of ALL guilty students.

COURSE INFORMATION

COURSE CODE AND TITLE:

ASD810S, Advanced Software Development

DEPARTMENT:

Software Engineering

PROGRAMME:

Bachelor of Computer Science Honors: Software Development

NOTIONAL HOURS:

Contact hours 45, Self-study hours 65, Assessment hours 40

NQF LEVEL AND CREDIT:

NQF level 8, 15 credits

COURSE DESCRIPTION:

This course will equip students with the necessary skills and knowledge of advanced software design, development, testing, and management techniques. Additionally, students should be able to critique the opportunities and challenges associated with developing software applications in the domain of Artificial Intelligence.

PRE-REQUISITES:

None

COURSE EQUIVALENCIES:

None

COURSE DELIVERY METHODS:

The course will be facilitated through the following learning activities:

- Lectures - this will present the underpinning theories of the course, and students are expected to deepen their knowledge and understanding of the covered subjects through guided self-study.
- Laboratory activities – this will be an avenue for students to apply and acquire practical skills based on the theoretical knowledge they have gained. Laboratory materials will be prepared with a clear indication of what the student is expected to achieve, and the problem statement and guidelines on how to approach the problem.

The following communication tools will be used in this course:

MS Teams, E-learning Platform (Moodle)

Course Format:

- 45 hours of face-to-face (or online via MS Teams) between students and lecturer
- 65 hours of out-of-class study time, to be used as follows:
 - 20 hours of laboratory (practical) session
 - 20 hours of reading prescribed material
 - 5 hours of sorting, completing, and annotating own lecture notes
 - 10 hours of working on assignments
 - 5 hours of consultations with lecturer and fellow students
- 15 hours of assessments:
 - 5 hours of completing online quizzes
 - 5 hours of tests
 - 5 hours of class and project presentations

EFFECTIVE DATE:

Monday 9 February 2026

LECTURER INFORMATION

Lecturer's name: Simon Muchinenyika

Email: smuchinenyika@nust.na

Office phone: 2072054

Office location: No. 201, Polyheights

Office hours: 07h30-12h30

Consultation hours: Thursdays 17h00

STUDENT READINESS

Technology & Equipment Readiness:

This course will only utilise already existing NUST equipment. Students will be using their own laptops for implementation and must ensure they have an appropriate backup.

Student Commitments and Contact Times:

1. Attend all classes and make appropriate notes. The presentations used in class and laboratory materials will be made available on Moodle or in the event of a glitch then on MS Teams. If a student fails to attend a class, it is their responsibility to catch-up on what was covered.
2. After every class, and before the start of the next class, the student should sort, complete and annotate own lecture notes, and select further reading in case any item was not understood on first attempt.
3. Student must fully utilise their notational hours on revising, practicing or/and completing homework for the course.
4. Consult recommended material to further broaden knowledge and understanding of the subject.

Course Resources:

Presentation slides, laboratory materials and additional materials will be made available on e-learning (Moodle).

Prescribed Reading:

Cowell, C., Lotz, N., & Timberlake C. (2023). *Automating DevOps with GitLab CI/CD Pipelines: Build efficient CI/CD pipelines to verify, secure, and deploy your code using real-life examples*. Packt Publishing, ISBN-13: 978-1803233000.

Richards, M., & Ford, N (2020). *Fundamentals of Software Architecture: An Engineering Approach*. O'Reilly Media.

Recommended Reading:

Singirikonda, E.K. (2024). *DevOps Automation Cookbook: Harness the Power of DevOps with 125+ Automation Recipes*. ISBN13 978-9355519061.

Blokdyk G. (2022). *Software Bots A Clear and Concise Reference*. 5STARCOOKS, ISBN-13 : 978-0655426035.

Kim, G., Humble, J., Febois, P., Willis, J., & Forsgrn, N. (2021). *The DevOps Handbook: How to Create World Class Agility, Reliability, & Security in Technology Organisations*. IT Revolution Press.

STUDENT LEARNING

Learning Outcomes:

On completing the course students will, through assessment activities, show evidence of their ability to:

1. Implement clean code based on best practices for software development.
2. Experiment with appropriate tools and techniques to enhance software development and delivery through collaborative efforts.
3. Optimise mobile software for mobile resource efficiency.
4. Investigate ways to integrate Artificial Intelligence into software development, considering the associated ethical considerations.
5. Develop AI-based applications essential to solving day-to-day challenges.

6. Analyse and discuss the limitations associated with AI-based solutions.

COURSE SCHEDULE:

Week Beginning	Topics	Tasks
9 February	Course Introduction Deliverables, Communication Channels, SWECOM	• Registration on MS Teams & E-Learning • Moodle Link
16 February	Software Development Fundamentals Programming, Version Control Systems, OS Terminal	<i>In-class Lab 1</i> • Git Lab • OS Terminal Commands
23 February	Testing Integration, Unit, Functional Tests Testing Platforms	• JUnit Tutorial • Test Frameworks • Testing & Verification
2 March	Architectural Design Systems Design Coupling & Cohesion, S.O.L.I.D.	• MVA: Minimum Vs Viable • Modularisation • Architecture Samples
9 March	Architectural Patterns Backend Architecture SOA, Service Mesh, Twelve Factor Apps	• Monolith • Microservices Example • Serverless
16 March	API Design API Styles Principles & Best Practices	<i>Mastery Learning Quiz 1</i> • REST dev • Flipped Classroom
23 March	Test	<i>Test</i>
30 March	Mid-Semester Break	Break
7 April	Containers Kubernetes Underlying Technologies	<i>Group Project Release</i> • namespaces • cgroups
13 April	Containers Docker LXC	<i>In-class Lab 2</i> • Language Translation App • Docker Lab
20 April	Deployment Patterns CI/CD Integration, GitOps, HelmCharts Canary Deployments, Rollbacks	<i>Mastery Learning Quiz 2</i> • GitHub Actions • Incremental Builds
27 April	Integration Patterns Streaming, Structured Outputs, Function Calling	<i>Supplementary Test</i> • Java Example • Class Group Exercise
5 May	Optimisation Profiling, Instrumentation, Monitoring Platform Specific Techniques	• Refactoring • Lab Exercises • Discussions
11 May	Group Project Presentation	<i>Group Project Presentation</i>

NOTE: Test and assignment dates are subject to change at the lecturer's discretion based on students' needs. Students will be notified in advance of any changes.

ASSESSMENT AND EVALUATION:

The semester mark is determined by continuous evaluation, which includes tests and individual assignments. The final course mark is the sum of the quizzes, test, and individual assignment.

Assessment	Weight
In-class Labs	20%
Quizzes	15%
Group Project	40%
Test	25%
Total:	100%

Minimum pass requirement for this course: 50%

COURSE POLICIES

General Academic Policies:

It is the student's responsibility to be familiar with and adhere to NUST's Policies. These policies can be found in NUST Prospectus or online at www.nust.na/prospectus.

Supplementary Policies:

If a student misses one of the tests for a valid, provable reason, one test will be considered instead of averaging the two. Take note that there are no supplementary assessments for quizzes and projects as the assessments are open for a long period of time.

All late submissions will not be accepted after the due date and time and thus will be marked as 0%.

DATE REVISED: 19 January 2026

FAILURE TO PAY FEES:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

IMPORTANT STUDENT SERVICES AT NUST

There are a variety of services which you can use at the NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development – Department: Students Services
- Writing Centre and student academic problems –Teaching and Learning Unit (CTL)
- Campus Health and Wellness Centre (CHWC) - Student Services'/ NUST Clinic

AUTHORISATION:

This course is authorised for use by:



Head of Department

06/02/2026

Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, (insert name), (Student number), hereby acknowledge that I have received this course outline for (insert course title and code), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of Student

Date